

Tim Randolph

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Harvey Mudd College
Department of Computer Science
McGregor CSC, Room 327
Claremont, CA 91711

APPOINTMENTS

Harvey Mudd College, Claremont, CA 2024-
Assistant Professor, Department of Computer Science

EDUCATION

Columbia University, New York, NY 2018-2024
Ph.D., Computer Science 2024
Thesis: “Exact Algorithms for Subset Sum and Subset Balancing Problems”
M.Phil., Computer Science 2022
M.S., Computer Science 2019
Advised by Xi Chen and Rocco A. Servedio

Williams College, Williamstown, MA 2014-2018
B.A., Computer Science (Highest Honors), Mathematics (Honors), Philosophy 2018
Thesis: “ (k, p) -Planar Graphs: A Generalized Planar Representation for Cluster Graphs”
Advised by William J. Lenhart

TEACHING EXPERIENCE

Harvey Mudd College, Claremont, CA

Instructor for CSCI 145: Advanced Topics in Algorithms S26
New course developed for S26. “Peer-to-peer” tutorial course in algorithms and complexity; heavy emphasis on presentation, communication, collaboration, and independent learning.

Instructor for CSCI 140 / MATH 168: Algorithms F24, S25, F25, S26, F26
Comprehensive undergraduate algorithms course.

Instructor for WRIT 1: Introduction to Academic Writing F25
Freshman course focused on developing academic writing skills.

Advisor for CSCI 183/184: Computer Science Clinic I & II F24/S25, F25/S26, F26/S27
Year-long project-based capstone course with 4-6 students. Advised Hawai’i Keiki Museum (F24/S25), UC Berkeley (F25/S26), and LS Power (F26/S27).

Columbia University, New York, NY

Instructor for COMS 3261: Computer Science Theory S21, S22, S23
Undergraduate automata, computability and complexity course.

Columbia CTL Teaching Development Program (Advanced Track) 2019-2022
Columbia CTL Teaching Observation Fellow 2019-2020
Columbia CTL Innovative Teaching Summer Institute (ITSI) 2019

JOURNAL AND CONFERENCE PUBLICATIONS[†]

My two research interests are in (1) algorithms (exact and parameterized algorithms, algorithmic combinatorics) and (2) teaching theoretical computer science.

**Peer-to-Peer Tutorials in Algorithms: Reflections on Scaling Up
Interactive Presentation-Based Learning** SIGCSE 2027

Tim Randolph

2027 ACM Technical Symposium on Computer Science Education

Beating Meet-in-the-Middle for Sumset Balancing Problems STOC 2026

Tim Randolph and Karol Węgrzycki

2026 Symposium on Theory of Computing

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Testing Sumsets is Hard ESA 2025

Xi Chen, Shivam Nadimpalli, Tim Randolph, Rocco Servedio, and Or Zamir

2025 European Symposium on Algorithms

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A Survey of Undergraduate Theory of Computing Curricula SIGCSE Virtual 2024

Ryan E. Dougherty (co-lead), Tim Randolph (co-lead), Tzu-Yi Chen, Jeff Erickson, Matthew Ferland, Dennis Komm, Jonathan Liu, Timothy Ng, Smaranda Sandu, Michael Shindler, Edward Talmage, and Thomas Zeume

2024 ACM Virtual Global Computing Education Conference

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**Parameterized Algorithms on Integer Sets with Small Doubling: Freiman's
Theorem, Subset Sum and k -Sum** ESA 2024

Tim Randolph and Karol Węgrzycki

2024 European Symposium on Algorithms

[View Online](#)

Participatory Governance in the Computer Science Theory Classroom SIGCSE 2024

Tim Randolph

2024 ACM Technical Symposium on Computer Science Education

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Subset Sum in $2^{n/2}/poly(n)$ Time RANDOM 2023

Xi Chen, Yaonan Jin, Tim Randolph, and Rocco Servedio

27th International Conference on Randomization and Computation

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[†]Publications in theoretical computer science list authors alphabetically by last name. Equal contribution is assumed unless otherwise stated. Conferences are primary publication venues.

Publications in computer science education list authors in order of contribution.

A Note on the Complexity of Private Simultaneous Messages with Many Parties ITC 2022
Marshall Ball and Tim Randolph
3rd Annual Conference on Information-Theoretic Cryptography
[View Online](#)

Average-Case Subset Balancing Problems SODA 2022
Xi Chen, Yaonan Jin, Tim Randolph, and Rocco Servedio
33rd Annual ACM-SIAM Symposium on Discrete Algorithms
[View Online](#)

Parallel Lotteries: Insights from Alaskan Hunting Permit Allocation MS 2021; EC 2021
Nick Arnosti and Tim Randolph
Management Science Vol. 68, No. 7 (Journal version)
22nd ACM Conference on Economics and Computation, as “Alaskan Hunting License Lotteries are Flexible and Approximately Efficient” (Conference abstract)
[View Online](#)

A Lower Bound on Cycle Finding in Sparse Digraphs SODA 2020; TALG 2022
Xi Chen, Tim Randolph, Rocco A. Servedio, and Tim Sun
ACM Transactions on Algorithms, Vol. 18, Issue 4 (Journal special issue)
31st Annual ACM-SIAM Symposium on Discrete Algorithms (Conference)
[View Online](#)

(k, p) -Planarity: A Relaxation of Hybrid Planarity WALCOM 2019; TCS 2021
Emilio di Giacomo, William J. Lenhart, Giuseppe Liotta, Timothy W. Randolph, and
Alessandra Tappini
Theoretical Computer Science, Vol. 896 (Journal special issue)
13th International Conference and Workshops on Algorithms and Computation (Conference)
[View Online](#)

Tight Bounds for $(t, 2)$ -Broadcast Domination on Finite Grids RHUMJ 2019
Timothy W. Randolph
Rose-Hulman Undergraduate Mathematics Journal, Vol. 20
[View Online](#)

Optimal (t, r) -Broadcasts on the Infinite Grid DAM 2019
Benjamin F. Drews, Pamela E. Harris, and Timothy W. Randolph
Discrete Applied Mathematics, Vol. 255
[View Online](#)

INVITED ARTICLES

Surveying Theory of Computation Education in the United States EATCS 2026
Ryan E. Dougherty and Tim Randolph
Bulletin of the European Association for Theoretical Computer Science No. 148
[View Online](#)

PREPRINTS

Integer Programming With Polyhedral Constraints, With Applications to Unbounded Subset Sum and Integer Programming with Small Doubling Preprint, 2026

Anubhav Dhar, Tim Randolph and Karol Węgrzycki

Exact Algorithms for Finding Sumsets Preprint, 2023

Tim Randolph

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INVITED TALKS, POSTERS, AND PANELS

RESEARCH PRESENTATIONS

“Beating Meet-in-the-Middle for Subset Balancing Problems” (talk)

Symposium on Theory of Computing (STOC), Salt Lake City, UT, 6/23/2026

“A Peer-to-Peer Tutorial Model for CS Theory and Algorithms” (talk)

Illinois Computer Science Summer Teaching Workshop, Virtual, 6/11/2026

“Pedagogy in Theory of Computing and Algorithms” (panel)

ACM Technical Symposium on Computer Science Education (SIGCSE), St. Louis, MO, 2/20/2026

“Exact and Parameterized Algorithms for Hard Problems” (talk)

Pomona College, Claremont, CA, 11/6/2025

“Subset Balancing: A Puzzle of Algorithmic (Additive?) Combinatorics” (talk)

University of California Irvine, Irvine, CA, 10/17/2025

“Exact and Parameterized Algorithms for Subset Sum Problems” (thesis defense)

Columbia University, New York, NY, 4/9/2024

“Designing Algorithms for Hard Problems: A Case Study” (talk)

Williams College Computer Science Colloquium Series, Williamstown, MA, 4/5/2024

“Experience Report: Participatory Governance in the Computer Science Theory Classroom”

(talk), *ACM Technical Symposium on Computer Science Education (SIGCSE)*, Portland, OR, 3/21/2024

“Algorithmic Approaches to Subset Sum (And Other Hard Problems)” (talk)

Harvey Mudd College, Pomona College, Amherst College, Bard College, Union College, 10/31/2023-11/25/2023

“Log Shaving for Subset Sum” (talk)

27th International Conference on Randomization and Computation (RANDOM 2023), Atlanta, GA, 9/12/2023

“The Complexity of PSM with Many Parties” (talk)

3rd Conference on Information-Theoretic Cryptography (ITC 2022), Boston, MA, 7/6/2022

“Average-Case Subset Balancing Problems” (talk)

31st Annual Symposium on Discrete Algorithms (SODA 2022), Virtual, 1/9/2022

“Parallel Lotteries: Insights from Alaskan Hunting Permit Allocation” (poster)

22nd Conference on Economics and Computation (EC 2021), Virtual, 7/21/2021

“Alaskan Hunting License Lotteries are Flexible & Approximately Efficient” (talk, poster)

DSI Financial and Business Analytics Center, New York, NY, 11/12/2019

15th Conference on Web and Internet Economics (WINE 2019), New York, NY, 12/10/2019

“The Case for Wasteful Allocation Mechanisms” (talk, poster)

1st INFORMS Workshop on Market Design, Phoenix, AZ, 6/28/2019

3rd Workshop on Mechanism Design for Social Good (MD4SG 2019), Phoenix, AZ, 6/28/2019

“(k,p)-planar Drawings of Cluster Graphs” (talk)

Williams College Summer Science Expo, Williamstown, MA, 8/11/2017

OUTREACH PRESENTATIONS

“Graduate School in Computer Science” (panel)

Harvey Mudd College Summer of CS, Claremont, CA, 7/8/2025

Southern California REU Symposium, Claremont, CA, 7/31/2025

“Navigating Graduate School in (Theoretical) Computer Science” (talk)

Columbia Emerging Scholars Program (ESP) Research Symposium, New York, NY, 4/12/2024

“Research and Exploration in (Theoretical) Computer Science” (talk)

Columbia Engineering Summer High School Academic Program (SHAPE), New York, NY, 8/11/2022

“Demystifying the Dissertation: Research in Theoretical Computer Science” (talk)

Columbia University Demystifying the Dissertation Seminar Series, New York, NY, 12/9/2020

“Research in Algorithms and Mechanism Design” (talk)

Columbia Emerging Scholars Program (ESP) Research Symposium, New York, NY, 11/20/2020

“Demystifying the PhD: Applying to PhD Programs” (talk)

Columbia University PhD Project Presentation Series, New York, NY, 11/18/2020

GRANTS AND AWARDS

Professional Development Network Grant, Claremont Colleges Faculty Development	2026
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Funding to create a community of practice for CS theory at the Claremont Colleges.

NSF REU Mentor, Harvey Mudd College.	2025
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Grant no. 2243941.

Internal Award for Summer Research, Harvey Mudd College	2025
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Michelman Award for Exemplary Service, Columbia University	2022
Columbia CS Department Service Award, Columbia University (3x)	2020, 2021, 2023
Sam Goldberg Prize, Williams College	2018
Elected Member, Sigma Xi	2018
Class of 1960s Scholar in Computer Science, Williams College (2x)	2017, 2018
Elected Member, Phi Beta Kappa (Junior Year)	2017
Class of 1960s Scholar in Cognitive Science, Williams College	2017

SERVICE

PROFESSIONAL SERVICE

Conference and Journal Reviews

ACM Conference on International Computing Education Research (ICER)
ACM Symposium on Theory of Computing (STOC)
ACM-SIAM Symposium on Discrete Algorithms (SODA)
ACM Special Interest Group on Computer Science Education Technical Symposium (SIGCSE)
ACM Transactions on Computing Education (TOCE)
Consortium for Computing Science in College: Southwestern Region (CCSC-SW)
European Symposium on Algorithms (ESA)
International Colloquium on Automata, Languages, and Programming (ICALP)
Symposium on Simplicity in Algorithms (SOSA)
Symposium on Theoretical Aspects of Computer Science (STACS)

Session Chair

ACM-SIAM Symposium on Discrete Algorithms (SODA 2022)

INSTITUTIONAL SERVICE

MAP-CS Working Group , Harvey Mudd College	2026-
<i>Long-term curriculum redesign committee participating in the 2026-2027 MAP-CS cohort.</i>	
Faculty Budget Committee , Harvey Mudd College	2025-
<i>Authored "HMC Faculty Guide to the Budget".</i>	
CS Curricular Redesign Committee Member , Harvey Mudd College	2024-2026
<i>Committee work during the redesign of the HMC introductory CS sequence. Participated in Capacity, External Compatibility, Mathematical Foundations, and Computational Thinking working groups.</i>	
PhD Student Representative , Columbia University	2022-2024
PhD Coordinator , Columbia University Emerging Scholars Program	2019-2022
Union Organizer , Student Workers of Columbia (UAW Local 2710)	2021-2022
Founding Organizer , Columbia Pre-Submission Application Review Program	2020-2021
Founding Organizer , Columbia Graduate Student Theory Retreat	2019-2021
Speaker , Columbia "Demystifying the Dissertation" Initiative	2020-2021

UNDERGRADUATE RESEARCH STUDENTS

SUMMER RESEARCH:

Mia Alexander (HMC '26), Riley Brown (University of Nebraska-Lincoln '27), Theo Julien (HMC '26), Xiaofeng Li (HMC '27), Uddant Patodia (HMC '28)

SEMESTER RESEARCH / INDEPENDENT STUDY

Uddant Patodia (HMC '28), Alice Yuan (HMC '28)

ADVISING AND MENTORSHIP

External Capstone Advisor , U.S. Military Academy at West Point	2024-2025
<i>Advised on “TheoryViz: A Visualizer Tool for Theory of Computing Concepts”. This capstone appeared as a poster in SIGCSE '25.</i>	
Mentor , Williams CS Alumni Mentorship Program	2022-2024
Mentor , Women in Science at Columbia (WISC)	2021-2024
Mentor , Lumiere Research Scholars Program	2022
Mentor , Barnard Better, Enhance, and Advance Research Series (BEARS)	2022
Advisor , Columbia Undergraduate Theory Seminar	2022